

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000011
Award Program	Pollinator Health Fund
Project Title	Can commodity crop weed management practices enhance bee abundance, diversity, and health on agricultural land?
Grant Start Date	01/15/2018
Grant End Date	01/14/2022
Total Grant Budget	\$241,800
Total FFAR Budget	\$120,900
Matching Funder(s)	University of Kentucky Research Foundation
Final Report Date	04/14/2022
Project Director/Principal Investigator Information	
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General Information

- 1.1. **Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.**

Lexington, KY KY-6

Franklin, KY KY-6

Owen, KY KY-6

Versailles, KY KY-6

Winchester, KY, KY-6

- 1.2. **How many new jobs were created because of this grant funding?**

6- 1 full time graduate student and 5 part time temporary employees.

- 1.3. **How many jobs were maintained because of this grant funding?**

1

- 1.4. **Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.**

No

- 1.5. **Has your organization undergone a recent name change? If so, please provide the new name of your organization.**

No

2. Scientific Report

- 2.1. **Public Abstract (up to 500 words): The abstract should be 'stand-alone'**



and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Bees feed on nectar and pollen collected from flowering plants. A shortage of floral resources, particularly in agricultural areas, is a major contributor to widespread declines in bee populations. This project investigated whether passive land management practices on croplands improve floral resource availability and bee survival during a critical, but understudied time of year, the late-winter/early-spring (LWES).

During LWES, most cropland is still dormant and between commodity crop cycles. These lands can be managed in a variety of ways, but one common strategy, especially in KY, is to leave fields fallow. Particularly on corn and soybean fields, this practice results in the abundant growth of small herbaceous “winter annuals”, which flower very early (e.g., late January) and bloom throughout LWES. Other management strategies exclude these winter annuals, eliminating the floral resource. Thus, determining the value of these winter annuals as floral resources for bees is critical to developing best management practices for dormant croplands.

From 2019-2021, we assessed whether honey bee colonies in close proximity to blooming winter annual plants showed enhanced LWES growth compared to colonies isolated from these floral resources (all colonies were kept on agricultural lands). We included 15 colonies across 4 sites in 2019, 32 colonies across 4 sites in 2020, and 20 colonies across 19 sites in 2021. We found that colonies on sites with a high abundance of blooming winter annual plants showed improved LWES growth in terms of weight gain and honey stores. We also found that variation in winter annual abundance significantly impacted the activity of a set of immune genes in forager bees collected from our colonies. This could reflect forager stress, because colonies isolated from winter annuals must travel farther to find food. Finally, landscape level analyses suggest that proximity to cropland improves colony growth during LWES, presumably because these lands contain more floral resources. We found no impact of proximity to natural areas during this time (which presumably contain blooming trees and shrubs). This result contrasts with studies at other times of year, illustrating the importance of assessing honey bee floral resource use throughout the active season.

We also assessed whether the presence of winter annual plants impacted wild bee communities on agricultural lands. Because many wild bees nest in the ground and could therefore nest on fields with blooming winter annuals, we also assessed whether soil glyphosate applications, common in these systems, impacted bee survival and growth. Sampling 8 sites that vary in winter annual abundance across three years, we found no evidence that blooming winter annual plants alter wild bee communities on croplands. We also found no evidence that glyphosate application impacts survivorship or growth of wild bees.

Improving bee survivorship is critical to food security in the U.S. Our findings suggest that leaving croplands fallow overwinter may provide critical food resources for honey bees. Surprisingly, beekeepers may benefit from placing their hives near

croplands over the winter. In contrast, cropland management practices in LWES do not appear to impact wild bees.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

The findings of this project could influence two major players in the agricultural community: corn and soybean growers and beekeepers. Although there is some evidence that honey bees improve soybean pollination, most corn and soybean growers do not rely on bees for pollination services. However, many express an interest in managing their lands in ways that have positive impacts on bee health (based on our communications with the Kentucky Soybean Board). Moreover, Kentucky contains many small, diversified family farms. Maintaining adequate pollinator communities is critical for growth of specialty crops on these farms.

Growers have a variety of choices for winter land management, each with different consequences for soil health and weed management. Our project is the first to suggest that pollination services (or at least bee health) is an important ecosystem service that may also be impacted by winter land management. Practices that increase the abundance of winter annual plants (e.g., avoiding fall herbicide applications, heavy tillage, and small grain cover crop planting) may improve the growth and productivity of nearby honey bee colonies. Although growers will consider their own needs when deciding how to manage their winter lands, our study provides information that may influence this decision.

This project also has important implications for beekeepers. Honey bees are portable, but good colony placement is critical to colony survivorship and growth. Beekeepers must balance practical considerations with food availability and pesticide risk. Increasingly, studies suggest that different land types have variable impacts on colony productivity throughout the year. For example, while blooming trees and shrubs on natural lands provide substantial food in mid-late spring, these areas become resource poor in the summer. In contrast, proximity



to croplands provides small herbaceous plants (abundant on agricultural field margins and pasturelands) that bloom during the summer. By investigating a poorly studied time of year (late-winter/early-spring), our study helps to complete our understanding of floral resource use throughout the active season, and inform beekeeper hive placement. On a local scale, fallow croplands are associated with greater colony growth, and this pattern extends to the landscape scale as well. Proximity to wooded areas did not influence growth, and pasturelands devoid of blooming herbaceous plants negatively influenced growth. Thus, placing colonies in landscapes with abundant cropland may be beneficial over the winter and during early spring, a critical pre-swarm period where hive growth influences beekeeper income. Strongly growing colonies can be split and sold, and colonies with strong early-season growth show enhanced growth potential for the remainder of the active season.

2.3. **Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:**

- a) **potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;**

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- b) **b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and**

Not applicable

- c) **c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.**

Not applicable



2.4. **What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)**

- 1) During late-winter/early spring (LWES), manipulate winter annual plant abundance in the vicinity of honey bee hives (within 200 m) and assess impacts on hive growth and worker immune system function. The **specific objectives** of this activity were to a) quantify the winter annual plant community around the site and identify plant types b) measure gains in hive weight, quantity of developing bees, and amount of stored food, and how these factors are impacted by winter annual plant abundance c) measure immune gene expression for foragers originating from hives with high versus low abundance of winter annual plants.
- 2) Assess variation in winter annual plant communities on fallow croplands during late-winter/early-spring and determine how winter annual plant abundance impacts the composition of the wild bee communities on croplands. The **specific objectives** of this objective were to a) use quadrats to sample winter annual plant communities on 200 m² plots composed of fallow cropland or planted with a small grain cover crop that excludes winter annual plant growth b) use bowl and vane traps to sample bee communities on plots c) sweep net bees visiting specific flowers to establish a plant-pollinator network, d) pin specimens and identify them to species.
- 3) Determine how a routine land management practice on croplands, spring glyphosate application, impacts the survival of ground nesting bees, because winter annual plant growth during late-winter/early spring may encourage these bees to nest on croplands. The **specific objectives** of this aim were to a) identify locations with aggregations of ground nesting bees and treat locations with a field-relevant glyphosate dose or a control once nesting was complete b) excavate nests to quantify the number of live individuals in plots treated with glyphosate versus on untreated control c) rear excavated larvae to adulthood in order to identify species d) weigh specimens and take body size measurements to determine sub-lethal impacts of glyphosate.

2.5. **Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)**

No



- 2.6. **What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)**

There were three **major activities** undertaken over the course of this research project. Each activity was completed each year of the project (three total years of data collection):

- 1) During late-winter/early spring (LWES), manipulate winter annual plant abundance in the vicinity of honey bee hives (within 200 m) and assess impacts on hive growth and worker immune system function. The **specific objectives** of this activity were to a) quantify the winter annual plant community around the site and identify plant types b) measure gains in hive weight, quantity of developing bees, and amount of stored food, and how these factors are impacted by winter annual plant abundance c) measure immune gene expression for foragers originating from hives with high versus low abundance of winter annual plants.
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For each major activity and specific objectives, we successfully collected data for three studies years over the course of this project. Here are the **major developments** and

findings.

1) We identified ~30-acre field sites that contained either high or low abundances of winter annual plants. High abundance sites were dominated by dormant corn and soybean crop fields, while low abundance sites were composed primarily of hay and pasture lands. Choosing sites in this way allowed us to maintain the high and low abundance of winter annual plants without site maintenance (grasses naturally outcompete the winter annual plants).

Each year, we took slightly different strategies to sample diverse sites with multiple honey bee colonies. In 2019, we took data from 15 hives distributed across four sites (two high and two low abundance). In 2020, we sampled 32 hives across the same four sites. In 2021, we sampled 20 hives across 19 unique sites in order to increase site-level replication, as recommended in response to our scientific talks on this topic. For each site, we sampled winter annual plant abundance and plant species identity throughout the late-winter/early-spring period (~Feb 25 – Apr 20, depending on year, sampling plant abundance 2-4 times per site per year). In 2019-2020 we collected hive inventory data and hive weights every 2 weeks, while in 2021, we collected hive inventory data at the beginning and end of the study period, and hive weight data continuously using hive sensors. In 2021, we also determined land cover percentages at 1, 3, and 7 km using ArcGIS and the National Landcover Database. This allowed us to assess how land cover at a larger spatial scale impacted colony growth.

In 2019 (N=2 high and N=2 low floral abundance sites, N=15 colonies across the four), starting brood quantity significantly predicted all final size measures (weight: $F_{1,13}=41.5$, $P<0.0001$; food: $F_{1,13}=10.0$, $P<0.01$; brood: $F_{1,13}=9.38$, $P<0.01$). Linear models including starting brood and site floral abundance category (high or low) as fixed effects, and site location as a random effect, showed significant effects of starting brood quantity, and a non-significant trend of an impact of floral abundance on final colony weight. No other factors were significant.

In 2020 (N=2 high and N=2 low floral abundance sites, N=31 colonies across the four), starting weight significantly predicted final weight and food (weight: $F_{1,29}=11.4$, $P<0.01$; food: $F_{1,29}=13.4$, $P=0.001$; brood: $F_{1,29}=2.07$, $P=0.16$), while starting brood was the best predictor for final brood ($F_{1,29}=5.6$, $P=0.025$). Linear models including starting weight (for final weight) or starting brood (for final brood quantity) and site floral abundance category (high or low) as fixed effects with site location as a random effect showed significant effects of the starting colony size measures, but not floral abundance.

In 2021, we altered our study design to increase the number of independent sites represented, giving 19 total independent data points. For this year, the only significant univariate relationship among growth measures was between starting brood and final colony weight (Linear Regression, $R^2 = 0.31$, $F_{1,17} = 7.7$, $P=0.01$). To assess the impacts of mean floral abundance per m² on colony growth parameters, we binned sites into quartiles based on floral abundance. We constructed linear models for final brood and food quantity and final weight that included mean floral abundance quartile, starting brood quantity, and their interaction as fixed effects. These models showed significant effects of mean floral

abundance on final food quantity and final weight, but no effect on final brood quantity (Table 1).

Table 1. In 2021, floral abundance at 200 m significantly predicted final colony weight and quantity of stored food. Sites were binned into quartiles on the basis of floral abundance, and those quartile values were included in the linear models.

		F	DF	P
Final food quantity	Starting brood	19.8	1	0.0006
	Floral abundance (200 m)	9.4	3	0.001
	Starting brood*Floral abundance (200 m)	2.3	3	0.13
Final brood quantity	Starting brood	1.2	1	0.3
	Floral abundance (200 m)	0.55	3	0.66
	Starting brood*Floral abundance (200 m)	0.72	3	0.6
Final weight	Starting brood	20.6	1	0.0005
	Floral abundance (200 m)	5.0	3	0.015
	Starting brood*Floral abundance (200 m)	1.9	3	0.19

Honey bees can forage 7-8 km from the colony. We used the National Land Cover Database and ArcGIS analyses to quantify landscape properties at 1 km, 3 km, and 7 km radii from colonies across 19 sites in 2021. We chose to focus our analysis on three major land cover types: two that are most relevant to winter annual abundance, cultivated cropland (typically high abundance of winter annuals) and hay/pasture lands (typically low abundance of winter annuals, but quite variable). We also assessed forest cover, summing across deciduous, evergreen, and mixed forest types for a "total forest" measure. Forested areas could contain attractive blooming trees and shrubs especially towards the end of LWES.

We first assessed whether the proportions of land cover for each of our three key features were correlated across spatial scales. We found significant correlations for each across 1 km, 3 km, and 7 km scales (the only exception was cultivated cropland, which was non-significantly correlated comparing 1 km and 7 km scales). Given strong land cover correlations identified using the NLCD, we also assessed relationships between land cover types at the 1 km scale and the 200 m scale. Rather than use the NLCD for the 200 m scale, we used our quadrat sampling data. For each quadrat sampled for floral abundance measures, we recorded surface type. From these records we categorized quadrats as cultivated crops (quadrats called "alfalfa", "soybean", "corn", "unknown crop fields"), total forest ("woods", regardless of forest type), and hay and pasture lands ("grass"). These categorizations do not exactly match the NLCD, but they reflect ground cover more precisely and at higher resolution. For example, a quadrat with grass could be on pasture lands or a roadside. Quadrat-based land cover estimates were significantly correlated with land cover proportions (cultivated crops, hay/pasture lands, total forest) at the 1 km scale.

Because land cover types both near the colony and at more distant spatial scales are correlated, we cannot meaningfully distinguish which spatial scale is most critical for colony growth in LWES. Because of the energetic costs of flight, near-colony



floral resources are typically prioritized, and foraging trips are often concentrated within about 1 km of the colony. It is also important to state that colony sites become increasingly non-independent at larger spatial scales (see below). Interestingly, though honey bees are capable of foraging at great distances, the land cover correlations across spatial scales in our data suggest that any colony growth patterns identified at small spatial scales are likely also predictive of outcomes at larger scales. This inference is important for beekeepers considering optimal colony placement in mixed landscapes.

Compared to our quadrat floral sampling method, the NLCD provides an indirect measure of exposure to winter annual plants at a lower and less precise resolution, but more comprehensive spatial scale compared. Thus, despite correlations in land cover type proportions, we evaluated whether land cover at 1 km is a significant predictor of colony growth measures. Though we cannot statistically differentiate impacts of the 200 m versus higher spatial scales on colony growth, the analyses at 1 km, at least to a degree, independently validate impacts of winter annual abundance on colony growth during LWES. We assessed Moran's I for final model residuals for each growth parameter to verify sites at this scale were statistically independent.

For all colony growth parameters, we built models to test the effects of cropland and total forest cover at 1 km (the two main sources of food during LWES) on colony growth. We verified that these two land cover measures are uncorrelated ($F_{1,17}=1.07$, $P=0.32$, $R^2=0.06$). We also determined which types of land cover, in combination with starting brood quantity, resulted in a model with the lowest AIC value for each growth measures to broadly assess impacts of land cover. For these analyses we considered 10 land cover types: developed land (summed across all intensities), deciduous forest, evergreen forest, mixed forest, total forest, hay/pasture, cultivated crops, and wetlands and open water.

Mirroring our findings for floral abundance at the 200 m scale, linear models including starting brood, cropland cover at 1 km, and total forest cover at 1 km as main effects showed cropland cover at 1 km explained significant variation in final food quantity, with a non-significant trend for final weight, and no relationship for final brood quantity. Total forest cover was not a significant main effect in any case. An assessment of AIC values for models containing different combinations of cropland, total forest cover, and starting brood as main effects showed a best final model that included starting brood and cropland cover for final weight and final food, and starting brood only for final brood. We assessed using univariate linear models whether cropland cover was correlated with any other land cover type, and found that it was negatively correlated with hay and pasture cover at 1 km ($F_{1,17}=12.3$, $P=0.003$, $R^2=0.42$). This result suggests that hay and pasture lands could negatively impact colony growth. However, in linear models for final weight, final food, and final brood that included starting brood and hay and pasture cover as main effects, hay and pasture cover did not explain significant variation in growth. Moreover, we compared AIC values for models containing starting brood and each other single land cover type. For final weight and final food, no other model came within two AIC units of the best model (cropland cover and starting brood as main effects). For final brood, all models were similar, suggesting no significant explanatory land cover main effect other than cropland. Thus, if hay and pasture cover impacts colony growth, it is likely an indirect effect, because hay and pasture lands are often associated with the absence of flower-

containing croplands.

For immune system function, we collected 320 individuals across 8 hives and two timepoints during LWES. Half of the individuals were given a yeast injection to simulate an infection. We then assessed fatbody gene expression patterns for an endogenous control gene and 5 target genes. Abaecin, Defensin-1, and Hymenoptaecin are genes that encode antimicrobial peptides. Vitellogenin and Insulin Receptor B are both indicators of nutritional status. We have completed processing all samples, and are currently analyzing the data to determine whether near-hive floral resources predict gene expression dynamics in foragers.

2) We sampled 8 200 m² research plots for winter annual plants and wild bee communities over three study years. Bee species identification and data analyses are on-going, but we have some general results to report. Overall, we trapped 2031 wild bees across three years. *Andrena* and *Lasioglossum* were the most commonly collected genera. We captured several species of bumble bee queens, which is significant because it emphasizes the fact that the timeframe of our study was during the delicate nest-founding period which has a high failure rate for bumble bees. Supplementing nutrition at this time could help queen bees be more successful. We also caught some floral specialist bees on our plots, which could indicate these bees are foraging more generally than initially thought.

Bee abundance varied greatly from year to year, largely due to variation in LWES weather. We collected relatively few bees in 2019 compared to 2020 and 2021. However, the abundance of bees captured overall suggests there are substantial communities on farmlands used in the study. The site rank in terms of abundance varied substantially from year to year. No site was consistently more likely to have wild bees present. The cause of these dynamics is unknown. We found a significant relationship between floral abundance on our plots and bee abundance in 2019, but no relationship in 2020 and 2021. 2019 was the harshest year in terms of weather, and the year we captured the fewest bees. Thus, nutrition could matter more in some years than others.

Analyses of plant communities are on-going. We found over 20 species of blooming winter annuals on our croplands. Purple deadnettle, chickweed, and speedwell were the most common species observed, making up the vast majority of plants on these fields. All three of these species are known to attract wild bees. Sites varied in the make-up of the plant communities. In general, blooming plant abundance increased over LWES, but these patterns varied year to year. We noticed diverse phenologic patterns across plant species represented in the study. Some showed steady increases in abundance over LWES, and some peaked early or in the middle of the study period. We collected data across research farms, and unlike the bee communities, weed communities were more similar for sites on a single farm.

3) We excavated a total of 19 plot pairs across all years of the study, 3 pairs in the spring of 2020 (Round 1; adults that grew up from individuals treated in June 2019), 8 pairs in the summer of 2020 (Round 2; larvae from individuals treated in June 2020), and 8 pairs in 2021 (Round 3; 74 larvae and 2 pupae from individuals treated in June 2021). There was a mean of 12.7 (s.d. =22.5) bees recovered from glyphosate-treated plots and 18.6 (s.d. =32.0) from control plots. Excluding 6 pairs of plots with zero bees recovered from both members of the pair, the mean number of bees per plot was 17.2

(s.d. =24.9) for glyphosate-treated plots and 27.2 (s.d. =35.8) for control plots.

Breaking down results by life stage at the time of excavation, in Round 1 (adults) we recovered an average of 40 individuals (s.d.=48.1) from glyphosate-treated plots, and 75 (s.d.=25.4) from control. Across Rounds 2 and 3 (almost all larvae, 6 pairs of plots with zero bees excluded), we recovered a mean of 12.2 bees (s.d.=12.1) from herbicide treated plots and 12.8 bees (s.d.=23.9) from control plots. We found no significant difference in the total number of bees recovered from glyphosate-treated versus control plots with data pooled across all years ($V=35.5$, $p=0.81$, $n=19$ pairs). Analyses separated by year and life-stage (adults in Round 1, pre-adults in Round 2 and 3) gave similar non-significant results.

Although we used very similar criteria to find, mark, and treat plots across all study rounds, we found substantial variation in the number of bees excavated from each plot, with no clear pattern based on location or time between plot marking and excavation. For example, we recovered the most bees per plot in Round 1, despite the shorter time between marking and excavation in Rounds 2 and 3. Similarly, despite ample bee traffic and confirmation of nesting activity, six plot pairs, 4 from Round 2 and 2 from Round 3, yielded zero bees. These bee-free plot pairs were distributed across five different parks, including the park we used successfully in Round 1. Thus, overall, although we find little evidence that glyphosate treatment impacts the survival of ground nesting bees, there are many sources of variation (some technical, though many biological) that may impact our ability to detect a treatment effect.

In Round 1, we excavated adult bees, which allowed us to assess patterns in species composition across paired plots. Our plot selection criteria involved monitoring *Andrena* and *Colletes* nests, and as expected, these two genera made up 68% of the adults excavated from the plots. However, *Lasioglossum*, which presumably nested outside of our observation window, was also very common (found at least once in all but one plot) and made up 32% of the excavated adults. Other excavated adults included one *Halictus*, one *Sphecodes*, and five *Nomada* total across all plots.

Because bee genera differ in size, and the effects of glyphosate treatment may vary across genera, we assessed the effects of glyphosate treatment on body weight and body size on a per-genus basis for *Andrena*, *Colletes*, and *Lasioglossum*. We excluded the rarer *Halictus*, *Sphecodes*, and *Nomada* from these analyses. We constructed linear mixed models with weight as the response variable. We included treatment as a fixed effect and plot pair as a random effect in all models. We also included sex as a fixed effect in the models for *Colletes* and *Andrena* because females are larger than males in these genera.

For *Andrena*, both treatment (Mean_{control} =10.4, S.D. =7.0 ; Mean_{herbicide}=21.8, S.D. =17.6 ; $\chi^2_{4,5}=6.2$, $p=0.01$) and sex (Mean_{female} =17.7, S.D. =14.9; Mean_{male}=9.9, S.D. =7.9; $\chi^2_{4,5}=18.3$, $p=1.9 \times 10^{-5}$) had significant impacts on weight. In contrast, for *Colletes*, sex was significant (Mean_{female} =65.1, S.D. =21.4; Mean_{male}=38.2, S.D. =12.8; $\chi^2_{4,5}=48.09$, $p=4.1 \times 10^{-12}$), but treatment was not (Mean_{control} =53.8, S.D. =22.5 ; Mean_{herbicide}=46.8, S.D. =20.6 ; $\chi^2_{4,5}=2.5$, $p=0.1$). Treatment did not significantly predict weight for *Lasioglossum* (Mean_{control} =7.5, S.D. =2.5 ; Mean_{herbicide}=8.8, S.D. =7.9 ; $\chi^2_{3,4}=1.0$, $p=0.3$). The differing responses to treatment by genus could be the result of several biological factors such as nesting behaviors and metabolism.

We also measured sclerotized body size using intertegumental distance, which is



a better estimate of size than wet weight because it is not impacted by water retention or metabolic rate (Cane 1987, Bosch and Vicens 2002). As above, we separated the bees by genus. Body size results mirrored the weight results above. In *Andrena*, both treatment ($\text{Mean}_{\text{control}} = 1645.9$, S.D. = 239.7; $\text{Mean}_{\text{herbicide}} = 1939.4$, S.D. = 354.6; $\chi^2_{4,5} = 6.4$, $p = 0.01$) and sex ($\text{Mean}_{\text{female}} = 1874.1$, S.D. = 284.9; $\text{Mean}_{\text{male}} = 1574.7$, S.D. = 275.7; $\chi^2_{4,5} = 32.3$, $p = 1.4 \times 10^{-8}$) had a significant impact on body size. In *Colletes*, sex had an impact on body size ($\text{Mean}_{\text{female}} = 3188.5$, S.D. = 355.4; $\text{Mean}_{\text{male}} = 2625.6$, S.D. = 251.5; $\chi^2_{4,5} = 62.7$, $p = 2.5 \times 10^{-15}$), but treatment did not ($\text{Mean}_{\text{control}} = 2813.2$, S.D. = 457.4; $\text{Mean}_{\text{herbicide}} = 2897.7$, S.D. = 374.3; $\chi^2_{4,5} = 0.6$, $p = 0.5$). Treatment did not have a significant effect on the body size of *Lasioglossum* ($\text{Mean}_{\text{control}} = 1434.3$, S.D. = 199.3; $\text{Mean}_{\text{herbicide}} = 1394.2$, S.D. = 199.3; $\chi^2_{3,4} = 1.5266$, $p = 0.2166$).

We took the wet weight of all Round 2 bees all at the same time (after completing all excavations), resulting in variable time between excavation and weighing among individuals. To account for this, we included excavation week in analyses of weight. 87 larvae died before we could weigh them, and so weights were unreliable due to mold, desiccation, or decay. We excluded these larvae from analyses. Most of the remaining living bees ($n = 40$) had expelled their gut contents and pupated at the time of weighing, while the rest were still larvae ($n = 26$). We analyzed live larvae and live pupae separately. For larvae, herbicide treatment did not significantly influence weight ($\text{Mean}_{\text{Control}} = 122.0$, S.D. = 44.1; $\text{Mean}_{\text{Herbicide}} = 98.7$, S.D. = 59.9; $\chi^2_{5,6} = 0.04$, $p = 0.85$), but excavation week did ($\text{Mean}_{\text{Week1}} = 91.2$, S.D. = 18.7; $\text{Mean}_{\text{Week2}} = 129.6$, S.D. = 46.8; $\text{Mean}_{\text{Week4}} = 53.9$, S.D. = 28.0; $\chi^2_{4,6} = 9.1$, $p = 0.011$). The relationship between excavation week and weight was not linear, suggesting development time or time under laboratory conditions, both potential sources of variation in final wet weight, do not fully explain the patterns. Because different plot pairs were excavated across the three excavation weeks, variation in weight may be due to species composition differences among sites.

We assessed the possibility that site, or park, may explain some of the weight variation by using ANOVA to compare linear mixed models containing park and treatment with plot pair as a random effect as above. However, the park from which larvae originated did not explain significant variation in weight ($\text{Mean}_{\text{Kirklevington}} = 53.9$, S.D. = 28.0; $\text{Mean}_{\text{Wildwood}} = 126.0$, S.D. = 46.1; $\chi^2_{4,5} = 0.1$, $p = 0.7$). We cannot fully evaluate the impact of species differences without identifying larvae.

We assessed Round 2 pupal weights using the same analyses as for larvae. There was no significant effect of the herbicide treatment ($\text{Mean}_{\text{Control}} = 128.1$, S.D. = 40.0; $\text{Mean}_{\text{Herbicide}} = 111.6$, S.D. = 39.7; $\chi^2_{3,4} = 1.9$, $p = 0.2$) or the excavation week ($\text{Mean}_{\text{Week1}} = 129.8$, S.D. = 33.7; $\text{Mean}_{\text{Week2}} = 120.4$, S.D. = 42.3; $\chi^2_{3,4} = 0.7$, $p = 0.4$) on pupal weights.

For Round 3 in 2021, we excavated plots over the course of 7 weeks, but we measured larval weights within 12 h of excavation. Thus, weight could vary as a function of development stage at the time of excavation, but not as a function of time spent in the lab. Using the same linear mixed models as for larvae in Round 2, we found no significant effect of herbicide treatment on larval weight ($\text{Mean}_{\text{Control}} = 119.7$, S.D. = 38.5; $\text{Mean}_{\text{Herbicide}} = 133.9$, S.D. = 43.0; $\chi^2_{3,4} = 1.9$, $p = 0.17$). Excavation week and park also gave non-significant results.

We attempted to rear the Round 2 larvae ($N = 137$) in the laboratory. However,

only 9 bees (4 from control and 5 from glyphosate treated plots) survived to adulthood, eclosing in early October. We froze these adults (N=9 total specimens) for later identification. Because we recorded the death date for all other individuals, we performed a survival analysis to determine if glyphosate treatment impacted laboratory survival. To do this, we constructed Kaplan-Meier survival curves and compared them using a log rank test. We also built a multivariate Cox proportional hazards model with glyphosate treatment included as an explanatory factor, and excavation week, life stage at excavation, and life stage at weighing as cofactors. We first analyzed both larvae and pupae together to determine if their life stage at the time they were weighed predicted survival. The Kaplan-Meier Survival Curve shows a lower, but non-significant, survival probability for pupae and larvae in the glyphosate group ($p=0.35$). We then used a univariate Cox proportional hazards model to determine the hazard ratio associated with herbicide treatment. This model shows a non-significant increase in hazard for herbicide treated bees ($z=0.8$, $HR=1.2$, $p=0.4$). The multivariate Cox proportional hazards model shows a significant reduction in hazard for bees excavated in Weeks 2 and 4 compared to Week 1, but non-significant increase in hazard to those excavated in Week 3 (Week2: $z=-4.2$, $HR=0.41$, $p=2.8 \times 10^{-5}$; Week 3: $z=0.03$, $HR=1.02$, $p=0.98$; Week 4: $z=-3.5$, $HR=0.46$, $p=.0003$). This change in hazard may be associated with the location of the nests or environmental conditions at the time of excavation.

Here are the **key outcomes** associated with each project activity:

- 1) Cropland appears to benefit honey bees during LWES because of the high level of blooming flowers on dormant fallow fields. Because land cover use is correlated across spatial scales, choosing hive placement that maximizes cropland close to the hive (200 m, easily assessed visually) is adequate to improve colony growth and influence worker bee immune system function in LWES. Land management practices that maximize winter annual growth on fallow fields (no-till, no fall herbicide treatment, no small grain cover crops) will significantly improve floral resource availability during a vulnerable time of year. Some growers may want to consider pollination services when they develop management plans for cropland, even in cases where crops do not themselves require pollination. It is easy to forget that these croplands nonetheless have substantial impacts on pollinators.
- 2) Wild bees are common visitors to plots with winter annual plants. Winter annual plant abundance predicted wild bee abundance in only the harshest weather year of the study, which could indicate that the impacts of these plants on nutrition is weather-dependent. *Andrena* and *Lasioglossum* make up the majority of specimens collected at our plots. We also collected several species of bumble bee queen, emphasizing the importance of LWES and perhaps winter annual plants for the risky nest founding process. We found over 20 species of winter annual plants on our sites. Chickweeds, speedwells, and purple dead nettle were the most common species. All of these are known to attract wild bees, emphasizing the potential value of winter annuals to the wild bee community.
- 3) Ground nesting bees can tolerate field relevant glyphosate applications in the late spring. If bees are attracted to fallow fields to nest, they will likely survive the typical herbicide



application practices that growers use to maximize commodity crop yield. Thus, we find no evidence that dormant lands serve as a “trap” that contains attractive flowers, but is otherwise harmful to wild bees.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

- A) Because this was a large, multi-year project, data management was an important component. All raw data sheets were scanned and backed up as a digital copy on the cloud. All data files were kept on the cloud, and accessible to different researchers in the study.
- B) Project design was vetted not only by the three co-PIs on this project, but also by Amanda Denny’s PhD advisory committee.
- C) We repeated our experiments across three years, each year improving the approach. For example, for activity #1, we increased our hive and site-level replication across the three years in order to make our results as robust and generalized as possible. Across all activities, we analyze and report data from all three years in order to deduce patterns that are consistent and shared.
- D) Data collection was performed by a number of students, faculty, and temporary work staff. We adopted an on-boarding protocol to ensure data collection was consistent across personnel.

2.8. List key stakeholders that could be served by results of this project.

Commercial, sideline, and hobby beekeepers
Specialty crop farmers
Corn and soybean growers
Diversified family farmers
The conservation-oriented public

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards



not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

This project had no formal requirement to disseminate information to the public. However, Dr. Rittschof and Amanda Denny provided data to communities through public outreach talks to beekeeper groups, master gardener groups, and students, totaling 22 presentations over the life of the project.

- 2.10. **Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)**

None to report

- 2.11. **What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)**

The project involved training a PhD student, Amanda Dunaway. This training involved a combination of weekly advisory meetings to discuss data, project progress, and project goal setting. These meetings also involve conversations about career goals and steps to meet those goals, including ways to develop the core competencies in experimental design, data analysis, and writing that are required of an independent scientist. Dr. Rittschof used her interactions with Amanda to identify strengths and weaknesses, and assigned weekly benchmarks to help her overcome difficulties. Over the course of the project, Denny was mentored through writing a USDA-AFRI predoctoral fellowship, and she submitted several small research award applications. She was given



feedback on talks that she gave at national and local scientific meetings. No formal IDP was used for this mentoring.

- 2.12. **Please indicate the number of undergraduate and graduate students, post- doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)**

Many undergraduate researchers assisted in the laboratory components of this project. Five undergraduate students assisted with molecular biology associated with activity #1. Three students also helped with wild bee specimen cleaning, pinning, and identification, as well as data entry.

3. Information Products

- 3.1. **Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.**

Scholarly publications, conference proceedings.

- 3.2. Please provide a list of citations for the information products produced during the project.

Conference proceedings

2020 ESA symposium: Pollinators and soybeans: Impacts, complications, and future directions (Remote). Impacts of corn and soybean winter land management on early-spring honey bee growth and wild bee activity

2020 Ohio State University Department of Entomology Seminar Series (Columbus, OH) Busy bees: integrative approaches to honey bee behavior and health



2020 Washington State University Dept of Entomology Seminar Series
(Remote) Wash your tarsi: linking social behavior and health outcomes in the honey bee

2020 Dunaway, A., Haramoto, E., Rittschof C.C. Weeds for bees: commodity crop weed management impacts on wild bee abundance. Entomological Society of America conference (Remote)

2021 Dunaway, A.S., Rittschof, C. Lost in the weeds: The impact of differing weed management strategies on bee communities. Presented at "Virtual North Central Branch Meeting", Entomological Society of America

2022 Dunaway, A.S., Rittschof, C. The role of flowering weeds as an early spring resource for native bees. Presented at "North Central Branch Meeting of the Entomological Society of America 2022", Entomological Society of America

Publications

Rittschof, C.C., Potter, D., Haramoto, E. (in preparation) Floral resources on dormant croplands improve honey bee colony growth during the pre-swarm winter and spring. To be submitted to: Agriculture, Ecosystems, and Environment (summer 2022)

Denny, A., Haramoto, E., Rittschof, C.C. (in preparation) Weed communities on fallow crop fields in the temperate east. To be submitted to: Agricultural Systems

Denny, A., Rehan, S., Haramoto, E., Rittschof, C.C. (in preparation) Winter management of dormant croplands does not impact wild bee communities. To be submitted to: Agricultural and Sustainable Food Systems

Denny, A., Haramoto, E., Rittschof, C.C. (in preparation) Glyphosate application does not impact ground-nesting bee survivorship. To be submitted to: Apidologie

Westwick, R.R. & Rittschof, C.C. (2021) Insects provide unique systems to investigate how early life experience alters the brain and behavior. **Frontiers in Behavioral Neuroscience** 15:660464.



Rittschof, C.C. & Nieh, J.C. (2021) Honey robbing: could human alterations to the environment change a rare foraging tactic into a maladaptive behavior? **Current Opinion in Insect Science** 45:84-90.

Carr, H.M., Palmer, J.H., Rittschof, C.C. (2020) Honey bee aggression: Evaluating causal links to disease-resistance traits and infection. **Behavioral Ecology and Sociobiology** 74:108.

Preston, S.R., Palmer, J.H., Harrison, J.W., Carr, H.M., Rittschof, C.C. (2019) The impacts of maternal stress on worker phenotypes in the honey bee. **Apidologie** 50:704-719.

Harrison, J.W., Palmer, J.H., Rittschof, C.C. (2019) Altering social cue perception impacts honey bee aggression with minimal impacts on aggression-related brain gene expression. **Scientific Reports** 9:14642.

3.3. **Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.**

Westwick, R.R. & Rittschof, C.C. (2021) Insects provide unique systems to investigate how early life experience alters the brain and behavior. **Frontiers in Behavioral Neuroscience** 15:660464.

Rittschof, C.C. & Nieh, J.C. (2021) Honey robbing: could human alterations to the environment change a rare foraging tactic into a maladaptive behavior? **Current Opinion in Insect Science** 45:84-90.

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Harrison, J.W., Palmer, J.H., Rittschof, C.C. (2019) Altering social cue perception impacts honey bee aggression with minimal impacts on aggression-related brain gene expression. **Scientific Reports** 9:14642.

- 3.4. **Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.**

Nothing to report

- 3.5. **Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).**

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

Nothing to report

- 3.6. **Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)**

Not applicable



- 3.7. **Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?**

All data files used in the analyses for our forthcoming manuscripts will be made publicly available.

4. Data Management

- 4.1. **Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.**

Wild bee specimens

Floral resource counts

Honey bee hive inventory data

Molecular data (qPCR readings)

Weights and counts of larval and pupal bees

- 4.2. **If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.**

Floral resource data correspond to sites on which wild bee and honey bee data were also collected

- 4.3. **Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.**

Relevant processed data files are included in a compressed folder.

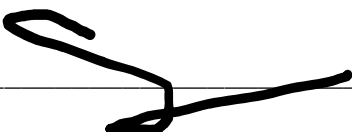


Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Clare C. Rittschof, PhD

PI Signature:  Date: Apr 14, 2022

Authorized Signing Official (ASO) Name: Kim C. Carter

ASO Signature:  Date: 4/14/2022

SMM
HCB